

ClassTrack Final Project Report

1. Project Overview

ClassTrack is a web-based learning management and tuition tracking system built for campus project use. The system supports two user roles:

- Admin: manages users, classes, monthly modules, paid enrollments, resources, and live class links.
- Student: browses classes, views monthly modules, unlocks paid content, joins live classes, and updates profile details.

The project combines backend PHP pages with MySQL data storage and a shared CSS design system. It also includes a static HTML prototype set used for UI mock/demo screens.

2. Project Goals

- Provide a simple class and monthly module management platform.
- Track student payment status per month.
- Restrict content access based on payment (paid vs locked).
- Allow admins to upload study resources and recordings.
- Give students a straightforward dashboard for learning access.

3. Technology Stack

- Backend: PHP (procedural pages)
- Database: MySQL (via PDO)
- Frontend: HTML + CSS + minimal inline JS
- Icons: Font Awesome CDN
- Runtime environment (expected): Apache + PHP + MySQL (for example XAMPP)

4. High-Level Architecture

The system is structured by role-specific modules:

- Public/Auth pages in project root:
 - index.php (landing page)
 - login.php (authentication)
 - logout.php (session termination)
- Admin module:
 - admin/dashboard.php
 - admin/users.php

- admin/courses.php
- admin/view_course.php
- admin/view_month.php
- Student module:
 - student/dashboard.php
 - student/course.php
 - student/month.php
 - student/profile.php
- Shared database connection:
 - includes/db.php
- Database definition:
 - database.sql
- Shared styling:
 - css/style.css
- Static prototype screens:
 - html/...

5. Database Analysis

5.1 Database Name

- class_track

5.2 Tables and Purpose

1. users
 - Stores all system users (admin and student).
 - Key fields: email (unique), password, role, full_name.
2. courses
 - Represents classes/grades (for example Grade 10 Science).
 - Linked to creator admin through created_by.
3. course_months
 - Represents monthly units/modules within a class.
 - Stores month name, year, live class link, and monthly fee.
4. enrollments
 - Links students to specific monthly modules.
 - Includes payment_status (pending/paid).
5. resources
 - Stores month-level educational content.
 - Resource type supports recording or document.

5.3 Relationship Summary

- users (admin) -> courses (created_by)
- courses -> course_months
- users (student) + course_months -> enrollments
- course_months -> resources

This creates a tuition-model structure where payment and access are tracked at month level rather than whole course level.

6. Functional Analysis (A-Z System Flow)

A. Landing and Login

- index.php presents a marketing-style entry page and login CTA.
- login.php validates credentials and starts role-based session.
- After login:
 - admin -> admin/dashboard.php
 - student -> student/dashboard.php

B. Session and Role Control

Every protected admin/student page checks:

- session user existence
- correct role value

Unauthorized users are redirected to login.

C. Admin Workflow

1. Dashboard
 - Shows counts of students, courses, and enrollments.
2. User Management
 - Admin can add users with role selection (student/admin).
 - Lists all users.
3. Class Management
 - Admin creates classes/grades.
 - Admin enters class detail page to manage months.
4. Month Management
 - Add month per class with year, fee, and optional live link.
5. Month Operations
 - Enroll student and mark payment as paid.
 - Add resources (recordings/documents) per month.
 - Update live link per month.

- View list of paid students and resources.

D. Student Workflow

1. Student Dashboard

- Lists all available classes.

2. Class View

- Shows all months for selected class.
- Displays lock/unlock state by payment status.

3. Month View

- Access allowed only for paid enrollment.
- Student can:
 - join live class if live link exists
 - watch recordings
 - download/view documents

4. Profile

- Student can update name and optionally password.
- Email is displayed but not editable.

7. UI/UX and Frontend Structure

- Single shared style file (css/style.css).
- Consistent layout blocks:
 - sticky header
 - card-based content areas
 - utility classes (container, spacing)
- Static HTML mirror exists under html folder:
 - used as prototype/demo version of PHP pages
 - includes simulated data and demo links

8. Setup and Execution Guide

8.1 Prerequisites

- PHP runtime
- MySQL server
- Apache web server
- Browser

8.2 Database Setup

1. Execute database.sql to create schema.

2. Ensure credentials in includes/db.php match local environment.

8.3 Run Application

1. Place project in web server root (for example htdocs).
2. Start Apache and MySQL.
3. Open index.php or login.php in browser.
4. Login with a user account stored in users table.

9. Strengths

- Clear separation of admin and student responsibilities.
- Simple and understandable database model for tuition logic.
- Payment-gated content flow implemented.
- Modular file organization by role.
- Basic responsiveness and clean UI consistency.

10. Conclusion

ClassTrack delivers the core objectives of a campus-level tuition and class management platform with role-based access, monthly payment-gated modules, and resource distribution. The architecture is clear and functionally complete for an academic project demonstration.